Classifying Aggressive Drivers for Better Traffic Signal Control

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***Abstract*—Traffic signal control, in a given single road intersection, primarily aims at reducing the average delay among a group of vehicles and avoiding crash accidents. Commonly, a driver's behavior is an essential cause of these accidents. There exist many solutions for automatic traffic signal control focusing on reducing average delay. However, to our knowledge, we are the first to integrate aggressive driving behavior classification with the traffic signal controller to enhance the safety and to reduce delays. Firstly, we present an artificial neural network as an aggressive driver behavior classifier trained using the benchmark Virginia Tech 100-car naturalistic driving study data. Secondly, we propose a modification to the popular Mamdani's fuzzy logic signal controller to accommodate for the integration of our aggressive behavior classifier. Multiple experiment sets were conducted to provide an indication to the effectiveness of our approach. The integration results yielded significant improvements at higher traffic flow volumes when compared against the baseline controller.**

Keywords—aggressive driver; classification; neural networks; support vector machines; fuzzy logic; traffic signal control

I. Introduction

According to the World Health Organization (WHO), the number of registered vehicles worldwide increased by 15% between 2010 and 2013 [1]. Increased traffic congestion in its worst case results in degraded use of the transportation infrastructure [2]. In addition, it has been estimated that around 1.2 million people die each year from road traffic crashes [3] with thousands more injured, some of them suffering long-term disabilities as a result [4]. Without a proper action to reduce these deaths, it is expected that traffic crashes would become the "7th leading cause of death by 2030" [5]. These statistics illustrate the urgent need for a solution regarding the road infrastructure in order to reduce congestion and increase the safety of the passengers and pedestrians.

Intelligent Transportation Systems (ITS) refer to a collection of techniques aimed at improving transportation systems using intelligent computing techniques. The goals of ITS include among others: reducing congestion, enhancing public safety, improving access to travel information, and reducing harmful environment impacts [6]. Traffic signal control, which is a part of ITS, aims to allocate right-of-way to a single or multiple non-conflicting traffic movements at a given time. One of traffic signal control's objectives is to reduce the average delay of a group of vehicles [7]. Three categories of traffic signal control exist: pre-timed, actuated, and adaptive control. In adaptive signal control, computational intelligence techniques are used to adjust signal timings by considering the conditions of the whole intersection [8].

Despite the intensive research and work done in Intelligent Transportation Systems, and particularly those related to traffic signal control, lack appears in the area of improving traffic delay and safety by incorporating the driver's behavior. Recognizing driver's behavior and early classification of aggressive behavior might be a new key to reducing intersection delays. In this paper, our aim is to improve traffic conditions through incorporating aggressive driver behavior classification within traffic signal control. To our knowledge, no previous work has been done which illustrates the importance of incorporating driver's behavior into traffic signal control. The first objective is to successfully build and test an aggressive driver behavior classifier, which would receive vehicle trajectory data and classify drivers into either being aggressive or non-aggressive. Secondly, the aggressive driver behavior classifier built shall be integrated with Mamdani's fuzzy logic traffic [9] controller in order to improve the intersection conditions with focus on vehicle delays.

The rest of this paper is organized as follows. Section II reviews some of the important literature on driver behavior classification and traffic signal control. Section III, then, presents the proposed architecture of both the driver behavior classifier and the modified fuzzy logic traffic controller. Section IV moves into discussing the experimental procedure, and Section V discusses the results. Finally, section VI concludes the paper.

II. Literature Review

Driver behavior classification was attempted in some research. Support Vector Machines and Bayesian Filters were used in [10] to classify "agents" into dangerous and harmless agents. An agent in their research referred to other drivers in the environment of the current driver (host). They defined a dangerous agent's goal as causing damage to the current vehicle or putting it in a risky situation, whereas a harmless agent's goal is to minimize time or distance. They identified the most critical features used in classification to be: the relative distance between the host and the suspicious vehicle, the heading of the suspicious vehicle relative to the host vehicle and the speed of the suspicious vehicle. Testing their classifier in 60 different

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scenarios, the classifier achieved a coverage of 100 percent and a precision of 77 percent. In [11] Support Vector Machine (SVM) classifier with a Bayesian Filter was tested to distinguish between the compliant drivers and violators based on the vehicle speed, acceleration and distance to intersection. They also used an HMM-based classifier which uses the expectation-maximization algorithm. The algorithms were successfully validated on naturalistic data and were shown to be effective in classifying drivers. The HMM had a higher true positive rate when the rate of false positives could increase past 10%, making it more suitable if detection accuracy is a major concern.

Other researchers focused on the analysis of vehicle trajectory patterns. Self-organizing feature map neural networks were used by [12] to learn the characteristics of normal vehicle trajectories. They represented trajectories using 2D image coordinates, together with first and second order motion information and a moving average smoothing. Training of the neural network was carried out on normal trajectories using video footage of a car park in normal conditions. The neural network can then be used in surveillance, whereby it marks object trajectories different from those encountered during the training phase as abnormal behavior. They achieved correct classification rate of 92% on 39 tested trajectories. Their major limitation lies in the difficulty in building prior knowledge into self-organizing networks and therefore the system was unable to differentiate between normal paths that have not been seen before and accurately suspicious trajectories. In [13] a whole vehicle trajectory was used as an input, unlike previous methods which depended only on individual flow vectors. They showed that the fuzzy self-organizing neural network outperforms both Kohonen self-organizing feature map (SOFM) and vector quantization in terms of both speed and accuracy. In addition, they achieved inspiring performance in anomaly detection (88.75% recognition rate) and activity prediction.

Traffic signals are used to assign right of way to one or more non-conflicting movements in a given intersection. Controlling traffic signals is essential in order to ensure reduced delay. Of the most widespread computational intelligence techniques used in traffic signal control is the fuzzy logic. Pappis and Mamdani [9] were the first to introduce the idea of using fuzzy logic to control traffic signal in a single intersection made of two one-way streets in order to reduce the average delay of vehicles. They used as their input variables: 1) the number of vehicles arriving at the approach having the right of way (also known as arrivals); and 2) the queue building up on the other approach. Their output variable was the extension in seconds to be given to the green signal. When compared against the conventional vehicle-actuated controller, their approach yielded an overall improvement in the delay by 10% ~ 20%.

In [14], the researchers targeted the implementation of a phase sequencer and developed their own simulation software simulating heavy traffic conditions. Results show more effective use of the intersection space (i.e. greater throughput) and less delay than pre-timed controller. However, they found out that the addition of the phase sequencer yielded no significant difference. In order to improve the performance of fuzzy logic systems, it could be beneficial to combine it with other computational intelligence techniques. Optimizing the membership functions usually results in better fuzzy logic systems. [15] used multi-objective genetic algorithms (MOGA) to find the set of optimal parameters of the membership function. Compared against the traditional vehicle-actuated control, lower delays were achieved (15 – 30% depending on the traffic conditions). Another approach introduced by [16] was to use genetic algorithms to improve some of the fuzzy rules in their rule set. Compared against fixed-time and actuated controllers, their approach produced the least total stop delay.

From the review of driver's behavior work, it can be concluded that research into classifying driver's behavior is successful. However, there is an evident lack of research on using feedforward and recurrent neural networks for classifying aggressive driver behavior, and this gives us a chance to test them for building aggressive driver classifier and comparing them with the performance of SVMs. Moreover, review of the work done on traffic signal control concluded that fuzzy logic control systems form the basis for most of the signal control systems. It is the concern of most traffic signal control systems to reduce traffic delays. It is also remarkable that no sufficient research has been carried out to test integrating driver behavior classification with traffic signal control. This provides us with an opportunity to incorporate driver behavior classification into traffic signal control in order to reduce delays.

III. Methodology

Integrating driver behavior classification within the framework of traffic signal control is challenging, as little work has been done in this area. Fig. 1 shows the overall proposed architecture, which consists of three main modules: vehicle data collection, either through video cameras and tracking program or through vehicle detectors placed on roads, driver behavior classification and finally fuzzy traffic signal control. The main focus in this research is on the two modules of driver behavior classification and fuzzy traffic signal control.

A. Driver Behavior Classification Module

The main aim of the driver behavior classification module is to test the feasibility of classifying aggressive and non-aggressive driving behavior. The input to the system is the event time series data obtained from the 100-car naturalistic driving study. Some preprocessing is firstly applied to the data to make it better suited for classification purposes. Lastly, data is passed to the final stage of classification. Three different classification algorithms were tested: feedforward neural networks, long short-term memory recurrent neural networks, and finally support vector machines.

B. Fuzzy Traffic Signal Control Module

Pappis and Mamdani's [9] traffic signal approach is the first traffic signal controller to use fuzzy logic. It is chosen in this paper for implementation, in order to test the effect of integrating the driver behavior classification module within it. Testing of new parameter integration within the basic approach should give an indication into the feasibility of integrating within more advances approaches. A model similar to Pappis and Mamdani's model was constructed and follows the following assumptions.

1. Arrival of vehicles at the intersection is considered to be random; that is, arrival times are uniformly distributed.

2. Total lost time of 10s/cycle is assumed.
3. No turning traffic.

1) Proposed Data Flow: Every second, data about the vehicles should be collected in order to use them for classification. This is achieved by keeping a trackedVehicles list, which contains the vehicles that are being tracked in the approach having right of way. Once a vehicle passes through the entrance detector, its data is collected and stored in the list. The three variables collected are: vehicle speed, acceleration and brake signal status. If the data collected for any vehicle exceeds 5 seconds, then the earliest entry is deleted, keeping the most recent 5 seconds' data. The flowchart in Fig. 2 shows what should typically happen every second before the controller intervention.

2) Proposed Fuzzy Linguistic Variable: Mamdani's fuzzy logic controller originally used two input arrays: alpha and alpha_dash. Alpha contains information about vehicles expected to cross the critical point in each of the next 10 seconds, while alpha_dash contains information about vehicles added to the queue in each of the next 10 seconds. Our additional linguistic variable that is added to the controller is Aggressiveness. This represents the ratio of aggressive drivers present in the arriving approach at a given point in time. Therefore, an additional controller input array (alpha_aggressive) is constructed. From this array, another array called (beta_aggressive) is constructed and is an accumulation of the values in alpha_aggressive array. A membership function similar to that used by Mamdani for the arrival variable is proposed for the aggressiveness variable and is shown in Fig. 3 . Also, a more than operation is defined on the variable.

3) Proposed Controller Operation: The proposed operation of the controller is also similar to Pappis and Mamdani's controller, with the addition of the aggressiveness ratio input. It can be summarized as follows.

1. The intervention of the controller takes place every 10s during each phase's effective green period. The value of 10s is chosen to be the same as Mamdani's controller.
2. The first intervention takes place just after the first 7s of the period. The value of 7s is also chosen to be the same as Mamdani's controller.
3. Decision taken: Length of extension of the effective green time for the phase having the right of way.
4. Control input parameters: three arrays corresponding to: Halted traffic (alpha_dash array), traffic having the right of way (alpha array), and ratio of aggressive drivers (alpha_aggressive array)
5. The fuzzy algorithm employed at each intervention for deciding the control action is represented by the union of five arrays (each array corresponds to a fuzzy rule, which is represented by a five-dimensional array).
6. Maximum of five interventions take place at: 7th, 17th, 27th, 37th, and 47th s. Thus, the maximum possible effective green time is 57 s.

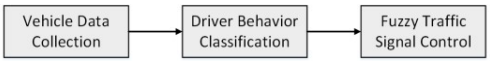


Fig. 1. Proposed Architecture.

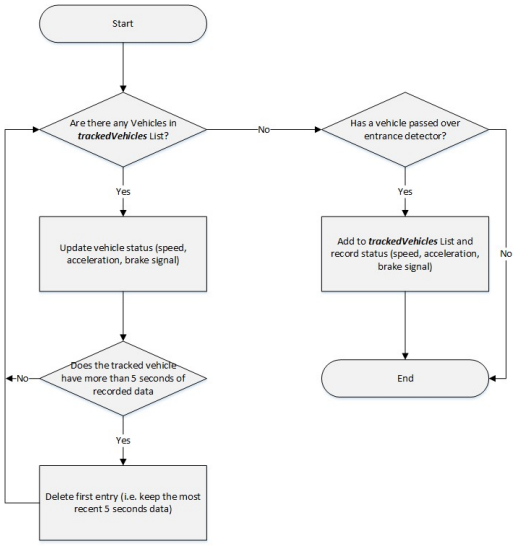


Fig. 2. Recording Vehicles' Statuses.

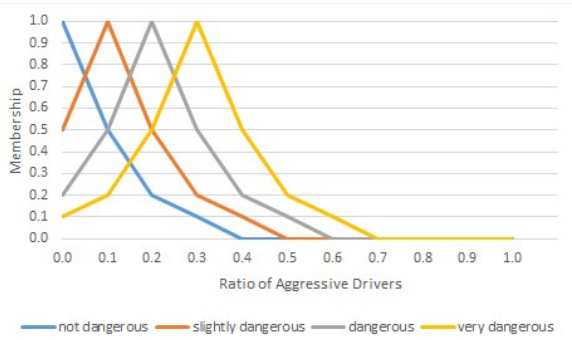


Fig. 3. Membership Functions of Aggressiveness Variable (Aggressiveness Fuzzy Membership 2).

7. At each intervention, the five rules are invoked (10 times).
8. At each intervention data is available for each of the next 10s.
9. Process of inferring the control action: For each successive time unit (1s) for the next 10s, data concerning vehicles crossing the critical point, vehicles added to the queue, and the ratio of aggressive drivers in the approach having right of way are used as input to the algorithm array in use.
10. The extension will be selected which corresponds to the maximum degree of confidence.

IV. Experimental Procedure

A. Dataset Description

The data used for the behavior classification module was the 100-car study data. The 100-car study is considered the first instrumented-vehicle study performed to collect large-scale naturalistic driving data. In the study, the drivers were not given any special instructions and no experimenter was present. 100 vehicles were used in the study and each vehicle's data was collected over 12 to 13 months [17]. This resulted in 2,000,000 vehicle miles and 43,000 hours of driving data. The publicly available part of the dataset contained a compilation of detailed video analysis for 68 crashes and 760 near crashes. Some preprocessing was performed on the time-series data, which consisted mainly of data aggregation, data reduction and normalization. Aggregation was achieved through time smoothing each second's 10 entries, thus producing a single entry per second rather than 10 separate entries. Data averaging aimed to reduce the data size, as well as reduce noisy data produced from the sensors. Then data reduction was carried out by reducing the initial feature set. Some of the fields are meaningless to include in the classification, such as the type of lane markings (double line, single line, etc.) and radar station ID, therefore the irrelevant fields were removed. The feature set, therefore, started with the following 10 features: vehicle speed, yaw rate, heading GPS, lateral acceleration, longitudinal acceleration, left distance to lane markings, right distance to lane markings, light intensity, brake on/off, and turn signal state. The samples were also filtered to remove any incomplete data (having less than 30 seconds of data prior to the event). After labelling and filtering the samples, there were a total of 788 trajectory records, of which 582 were labelled non-dangerous and 206 were dangerous. Finally, normalization was carried out to avoid dependence on the measuring unit, and to give each attribute an equal weight. Each attribute was scaled to fall in the range [-1, 1]. In (1), value x_i, of attribute A is mapped to x_i' in the range [-1, 1], where min_A and max_A are the minimum and maximum values of attribute A. In classification, data was randomly divided into training (70%), validation (15%) and testing (15%) sets.

$$x_i' = \frac{2*(x_i - min_A)}{max_A - min_A} - 1 \quad (1)$$

For the fuzzy traffic signal control module, Simulation of Urban MObility (SUMO) was used to simulate traffic flow and traffic signal control. SUMO is an open-source microscopic vehicular traffic simulator developed at German Aerospace center [18]. It also provides an interface for traffic signal control called Traffic Control Interface (TraCI), which allows to "retrieve values of simulated objects and to manipulate their behavior on-line" [19]. A simple intersection with two approaches was used: North – South and East – West, and 16 different configurations of mean arrival rate were simulated as described in [9]. Some form of aggressive driving had to be exhibited by the vehicles produced in the simulation environment. A formal definition of aggressive driving was offered by Tasca in [20] as the driving behavior that is "deliberate, likely to increase the risk of collision and is motivated by impatience, annoyance, hostility and/or an attempt to save time", such as tailgating, improper passing, running red lights, driving at very high speeds, etc. In order to map aggressive driving behavior in SUMO, firstly 12 driver types were defined as follows: 1) Young well-behaved male, 2) Young well-behaved female, 3) Young aggressive male, 4) Young aggressive female, 5) Middle-aged well-behaved male, 6) Middle-aged well-behaved female, 7) Middle-aged aggressive male, 8) Middle-aged aggressive female, 9) Elder well-behaved male, 10) Elder well-behaved female, 11) Elder aggressive male, and 12) Elder aggressive female. Secondly, several car-following variables were modified in each driving category to map its degree of aggressiveness. The car-following model used in our SUMO simulations was developed by Krauß [21] and contains a set of parameters which could map the meniontedaggressive behaviors. TABLE I shows the vehicle and car following parameters used to map the aggressive behaviors and Fig. 4 shows the values used for each driver type.

B. Performance Evaluation Measures

To evaluate the effectiveness of our approach, two main modules should be evaluated: 1) driver behavior classification; and 2) fuzzy logic controller enhancement. In order to compare the different classification models experimented in this research, the Receiver Operating Characteristics (ROC) curve is used. According to [22] an ROC curve is a "technique for visualizing, organizing and selecting classifiers based on their performance". For the purpose of this research the positive class is the aggressive behavior, while the non-aggressive behavior is the negative class, with the probability of aggressiveness used as the threshold. ROC curves act as a good visual tool comparing the different classifiers' performance; however, it would be easier if there is a single scalar number comparing performance. This scalar value can be extracted from the ROC and is the area under the curve (AUC). A classifier with a higher AUC is expected to have a better average performance. For the second module of traffic signal controller, the intersection delay is used as the performance evaluation measure. According to [9], intersection delay is defined as the number of seconds a vehicle stayed inside the intersection. In order to compare delay from our approach with the other approaches, different configurations of traffic flow and drivers' behavior are used and delay calculated for each configuration. The results are then compared with the basic approach to traffic control.

C. Driver Behavior Classification Module Experiments

This set of experiments aim to build a good aggressive driver classifier using the preprocessed 100-car study data. Three classification algorithms were tested: Feed forward Artificial

TABLE I. Car-following parameters mapping aggressive driving behavior

Vehicle/Car-following Parameter	Aggressive Driving Behavior Mapped
Acceleration (m/s^2)	The higher, the more aggressive the driver.
Deceleration (m/s^2)	The higher, the more aggressive the driver.
Sigma (Driver imperfection between 0 and 1)	Imperfect drivers would cause more intersection delays, and therefore the lower the value (i.e. more imperfect), the more aggressive the driver.
minGap (m)	Tailgating. The less the gap between the lead vehicle and the current vehicle, the more tailgating effect produced.

maxSpeed (m/s)	The higher the speed, the more aggressive the driving is.
speedDev (Deviation from vehicle speed)	The higher the speed deviation the more aggressive the driving behavior is.
tau (s) (The driver's desired minimum time headway)	Driver's reaction time. It is lower in more aggressive drivers.

Type		acceleration ($\frac{m}{s^2}$)	deceleration ($\frac{m}{s^2}$)	sigma	minGap (m)	maxSpeed ($\frac{m}{s}$)	speedDev	tau (s)
Young	well-behaved male	2.7	4.7	0.4	2.5	25	0.3	1
	well-behaved female	2.6	4.6	0.4	2.5	25	0.3	1
	aggressive male	3.2	5.5	0.3	0.9	35	0.1	1
	aggressive female	3	5.3	0.3	1	35	0.2	1
Middle-aged	well-behaved male	**2.6**	**4.5**	**0.5**	**2.5**	**23**	**0.5**	**1.5**
	well-behaved female	2.5	4.4	0.5	2.5	23	0.5	1.5
	aggressive male	3	5	0.4	1.1	28	0.3	1.3
	aggressive female	2.8	4.9	0.4	1.2	28	0.4	1.4
Elder	well-behaved male	2.5	4	0.6	2.5	23	0.6	1.9
	well-behaved female	2.4	3.9	0.6	2.5	23	0.6	2
	aggressive male	2.6	4.5	0.5	2.1	25	0.4	1.7
	aggressive female	2.5	4.3	0.5	2.2	25	0.5	1.8

Fig. 4. Vehicle and car-following properties of each driver type.

Neural Networks (ANN), Long Short-Term Memory Recurrent Neural Networks (LSTM-RNNs) and SVM. Several experiments were carried out testing different data configurations to reach to the best classification performance. The different configurations tested the effect of data averaging, reducing the number of features, and using different time spans as shown in TABLE II. These experiments were repeated for each of the three classification algorithms. The initial feature set consisted of the features that could be retrieved from the simulator, because further experiments of the controller build on this module.

D. Aggressive Driving Behavior Classifier and Fuzzy Logic Controller Integration

In this set of experiments, the aim is to test the feasibility of integrating the driver behavior classification module with a fuzzy logic controller. Two sets of experiments are carried out: 1) Integration of the classifier and the controller; and 2) Effect of different membership functions. The first set of experiments aim to test the proposed approach to reducing intersection delays through incorporating aggressive driver behavior classification into fuzzy logic controller operation. The trial and error method was followed to reach to the final rule set.

The second set of experiments intend to test different membership functions and their effect on the updated controller's performance. This is important in order to make sure that the membership function chosen gives the best results. Three different ranges of membership function were tested as shown in Fig. 3, Fig. 6 and Fig. 7. In these membership functions, the peaks of the fuzzy sets are being shifted to the right.

V. RESULTS

Comparison of the Area Under the Curve (AUC) of the 27 classification experiments (Fig. 8) shows that the best results were achieved in experiment 9 which used feedforward neural network on averaged data, with 3 features per second (vehicle speed, longitudinal acceleration and brake on/off signal) and a timespan of 5 seconds. The configuration of this neural network is shown in Fig. 5. Fig. 9 shows the delay after integrating the aggressive driver behavior classifier within Mamdani's fuzzy logic controller. It can be observed that the new controller results in lower delays in most of the traffic flows, with the enhancements more significant at higher traffic flows. The enhancement in terms of lower delays can reach up to 30%. When comparing the different membership functions, the results in Fig. 10 show that membership function 1 gave higher delays when compared against membership functions 2 and 3, which gave very near delay results. This indicates that higher value membership functions are more suitable for the proposed controller. The membership function which is chosen for the controller is the second one.

VI. CONCLUSION

The paper discussed two main contributions. Firstly, an aggressive driver behavior classifier was proposed. Previous work attempted classifying driver behavior from vehicle trajectory; however, there was an evident lack of research on using feedforward and recurrent neural networks for classifying aggressive driver behavior. Therefore, this gave us a chance to test two different kinds of neural networks under different data and network configurations and compare them with the performance of SVMs. The 100-car study dataset was used to build such classifiers, and to our knowledge, these are the first experiments to classify driver behavior using this data. Best results were achieved when using a feedforward neural network on averaged data, with 3 features per second (vehicle speed, longitudinal acceleration and brake on/off signal) and a timespan of 5 seconds. Secondly, literature review revealed an apparent deficiency testing the integration of driver behavior classification with traffic signal control. This gave us an opportunity to incorporate the aggressive behavior classifier within the fuzzy logic controller. Experiments showed that the proposed architecture was successful in reducing vehicle delays. It is not expected that this system would allow drivers to pretend being aggressive in order to pass the traffic light, as the traffic light behavior is totally unexpected. In the future, more experiments can be conducted to study the effect of the proposed controller under different traffic conditions.

TABLE II. CLASSIFICATION EXPERIMENTS CONFIGURATIONS

Exp #	Time span (sec)	Num Feat/ sec	Veh. Speed	Brake Signal on/off	Long. Acc.	Turn Signal State	GPS Heading
1	30	5	✓	✓	✓	✓	✓
2	30	4	✓	✓	✓	✓	×
3	30	3	✓	✓	✓	×	×
4	30	2	✓	✓	×	×	×
5	25	3	✓	✓	×	×	×
6	20	3	✓	✓	×	×	×
7	15	3	✓	✓	×	×	×
8	10	3	✓	✓	×	×	×
9	5	3	✓	✓	×	×	×

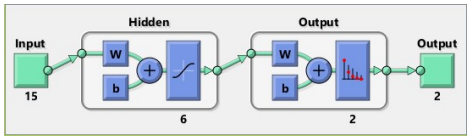


Fig. 5. Configuration of NN giving best results.

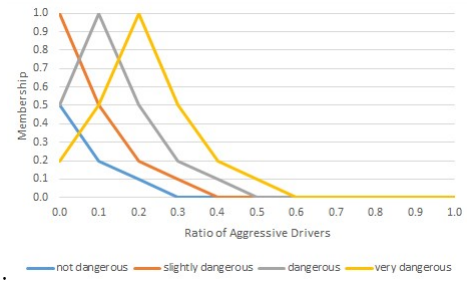


Fig. 6. Aggressiveness Fuzzy Membership 1.

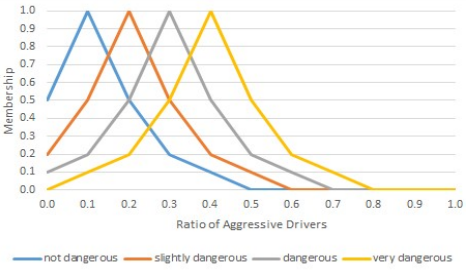


Fig. 7. Aggressiveness Fuzzy Membership 3.

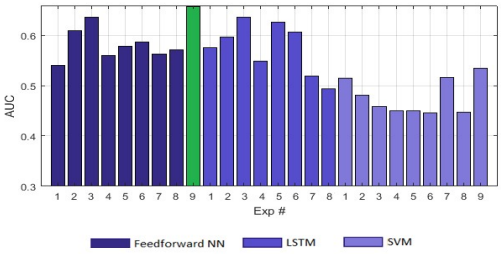


Fig. 8. AUCs for Classifier Experiments.

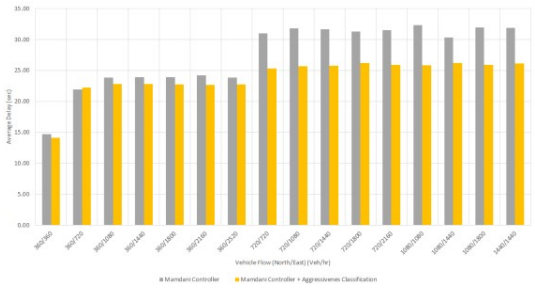


Fig. 9. Average Vehicle Delay from Proposed Controller vs Mamdani's Controller.

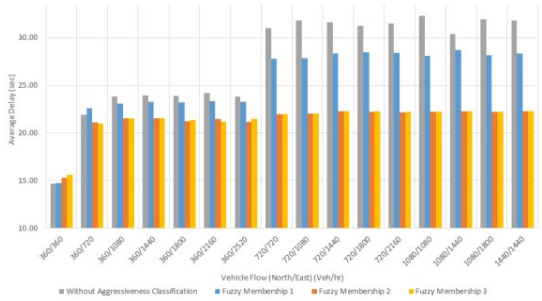


Fig. 10. Effect of Changing Aggressiveness Membership Function.

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